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LINUX SYSTEM PROGRAMMING (LSP)

TASK – 6

Privilege Levels Essay:

Write a detailed essay explaining Linux's privilege levels, focusing on Kernel Mode and User Mode.

Include examples of operations permitted at each level.

A- In Linux, as in most modern operating systems, privilege levels are a foundational concept that enforces security and stability. These levels determine the level of access a running process has to critical system resources. The two main privilege levels in Linux are Kernel Mode and User Mode. Each mode serves a purpose and is crucial to the operating system functioning.

Kernel Mode

Kernel Mode (also known as privileged mode or superuser mode) is the highest level of privilege in Linux. When the system is operating in Kernel mode, it has unrestricted access to all hardware and system resources. This mode is reserved for the Linux kernel - the core of the operating system - which performs low-level tasks and manages hardware operations such as memory management, scheduling, and device I/O.

Characteristics of Kernel Mode -

- Direct access to hardware devices.
- Execution of any CPU instructions.
- Unrestricted access to memory.
- Ability to manage processes and system calls.

Examples of Operations in Kernel Mode

- 1 Managing Memory: Allocating and deallocating memory pages.
- 2 Interfacing with Hardware: Communicating with device drivers, such as reading data from a disk or sending packets through a network interface.
- 3 Handling System Calls: Executing requests from user applications like `read()`, `write()`, or `fork()`.
- 4 Scheduling Processes: Choosing which process to run next based on priority and resource availability.

Kernel mode code must be carefully written because a bug or vulnerability here can crash the entire system or create serious security risks.

User Mode

User mode is a restricted processing mode designed for user applications. Code running in user mode cannot directly access hardware or reference kernel memory. If a program in user mode requires access to system resources, it must request it via system calls, which safely transition execution to kernel mode.

Characteristics of User Mode:

- Limited access to system resources.
- Cannot execute privileged CPU instructions
- Cannot directly access hardware or kernel memory
- Must use system calls to request kernel services.

Examples of Operations in User Mode:

- 1 Running applications: Any standard applications like a web browser, text editor, or media player runs in user mode.
- 2 File Manipulation via System Calls: Using high-level functions like `fopen()` in C, which internally invoke system calls to access files.
- 3 Requesting data from devices: A user program may request to read a file from the disk, but the actual I/O is handled by the kernel.